

# Part II V3 in Lamport notation

Construction of the continuum torus Hamiltonian, trace class, periodic Feynman–Kac formula, and the gap

Fresh reconstruction of `part_ii/v3_continuum_package.tex`

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The continuum package is reconstructed below without importing a pre-existing  $P(\varphi)_2$  Hamiltonian. The only printed input is Simon’s four-result second-quantization cluster, checked in the accompanying citation-evidence dossier. The periodic cutoff estimate that V3 states only by analogy is supplied explicitly in Steps  $\langle 3 \rangle 8$ – $\langle 3 \rangle 10$ .

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## 1 Data, exact target, and imports

Fix  $L, m > 0$ ,  $\lambda \geq 0$ , and  $a, b \in \mathbb{R}$ . Put

$$\Gamma_\infty = \frac{\pi}{L}\mathbb{Z}, \quad \gamma(k) = (m^2 + k^2)^{1/2}, \quad \mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z}). \quad (0.1)$$

Whenever a periodic time interval is used, fix  $t \in [t_0, T]$  and impose

$$t_0 L \geq 1. \quad (0.2)$$

The gate ultimately uses  $(a, b) = (0, 0)$ . Allowing arbitrary  $(a, b)$  here makes explicit that no continuum counterterm is being hidden.

Let  $(e_j)_{j \geq 1}$  be the real orthonormal Fourier basis

$$e_0 = (2L)^{-1/2}, \quad e_k^c = L^{-1/2} \cos(kx), \quad e_k^s = L^{-1/2} \sin(kx) \quad (k > 0), \quad (0.3)$$

and let  $\gamma_j$  be the corresponding value of  $\gamma$ . On

$$(Q, \mu) = \left( \mathbb{R}^{\mathbb{N}}, \bigotimes_{j \geq 1} \mathcal{N}(0, 1) \right) \quad (0.4)$$

write the coordinate variables as  $\xi_j$  and define

$$\varphi^{(K)}(x) = \sum_{|k_j| \leq K} \frac{\xi_j}{\sqrt{2\gamma_j}} e_j(x), \quad c_K = \mathbb{E}_\mu[\varphi^{(K)}(x)^2] = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{2\gamma(k)}. \quad (0.5)$$

The free Hamiltonian on  $L^2(Q, \mu)$  is

$$H_0 = d\Gamma(\gamma), \quad e^{-sH_0} = \Gamma(e^{-s\gamma}), \quad Z_0(s) = \text{Tr } e^{-sH_0} = \prod_{k \in \Gamma_\infty} (1 - e^{-s\gamma(k)})^{-1}. \quad (0.6)$$

Its bounded smooth cylinder core is

$$\mathcal{D} = \{F(\xi_1, \dots, \xi_n) : n < \infty, F \in C_b^\infty(\mathbb{R}^n)\}. \quad (0.7)$$

The following are the only external statements used.

**Imported assumption 1.1** (Simon second-quantization cluster). For a real-Hilbert-space contraction  $A$ , the Gaussian second quantization  $\Gamma(A)$  is positivity preserving; it is an  $L^p$  contraction for every  $1 \leq p \leq \infty$ ; it is positivity improving if  $\|A\| < 1$ ; and

$$\|\Gamma(A)f\|_q \leq \|f\|_p \quad \text{if } 1 < p \leq q < \infty, \quad \|A\| \leq \sqrt{\frac{p-1}{q-1}}. \quad (0.8)$$

These are Simon, Theorem I.12, Corollary I.15, Theorems I.16 and I.17. The exact pages are reproduced in the evidence dossier.

The elementary bounded-perturbation Trotter product formula and the spectral theorem are used in the forms proved in the model-sheet reconstruction. All remaining inputs below are results already proved in the V1 and repaired V2 Lamport modules.

## 2 Interaction and sharp-time convergence

$\langle 1 \rangle 1$  Bound the cutoff variance.

$\langle 2 \rangle 1$  For  $K \geq m$ ,

$$c_K \leq C_c(m, L) [1 + \log(1 + K/m)]. \quad (1.1)$$

*Because:* Pair the sine and cosine modes and compare the decreasing summand with the integral:

$$\sum_{0 < k \leq K} \frac{1}{\sqrt{m^2 + k^2}} \leq \frac{L}{\pi} \int_0^K \frac{du}{\sqrt{m^2 + u^2}} + \frac{1}{m} = \frac{L}{\pi} \text{arsinh}(K/m) + \frac{1}{m}.$$

Since  $\operatorname{arsinh} x \leq 1 + \log(1 + x)$ , (1.1) follows.

$\langle 2 \rangle 2$  This proves  $\langle 1 \rangle 1$ . □

$\langle 1 \rangle 2$  Define and bound the mode-cutoff interaction.

For  $K \in \Gamma_\infty$ , set

$$V^{(K)} = \lambda \int_{\mathbb{T}_L} \left[ :\varphi^{(K)}(x)^4:_{c_K} + a:\varphi^{(K)}(x)^2:_{c_K} + b \right] dx. \quad (1.2)$$

$\langle 2 \rangle 1$  For  $u \in \mathbb{R}$  and  $c \geq 0$ ,

$$:u^4:_c + a:u^2:_c + b = (u^2 - 3c)^2 + a(u^2 - 3c) - 6c^2 + 2ac + b. \quad (1.3)$$

*Because:* Expanding the right side gives  $u^4 + (a - 6c)u^2 + 3c^2 - ac + b$ , which is  $(u^4 - 6cu^2 + 3c^2) + a(u^2 - c) + b$ .

$\langle 2 \rangle 2$  Consequently,

$$:u^4:_c + a:u^2:_c + b \geq - \left( 6c^2 + 2|a|c + \frac{a^2}{4} + |b| \right). \quad (1.4)$$

*Because:* For  $v = u^2 - 3c$ ,  $v^2 + av = (v + a/2)^2 - a^2/4 \geq -a^2/4$ ;  $2ac + b \geq -2|a|c - |b|$ .

$\langle 2 \rangle 3$  Thus

$$V^{(K)} \geq -b_K, \quad b_K := 2L\lambda \left( 6c_K^2 + 2|a|c_K + \frac{a^2}{4} + |b| \right) \leq C[1 + \log(1 + K/m)]^2. \quad (1.5)$$

$\langle 2 \rangle 4$  If  $\lambda > 0$ ,  $V^{(K)}(\xi) \rightarrow +\infty$  as  $|\xi| \rightarrow \infty$  in the finitely many retained coordinates.

*Because:* Cauchy–Schwarz on a space of volume  $2L$  and Parseval give

$$\int_{\mathbb{T}_L} (\varphi^{(K)})^4 dx \geq \frac{1}{2L} \left( \int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx \right)^2, \quad \int_{\mathbb{T}_L} (\varphi^{(K)})^2 dx = \sum_{|k_j| \leq K} \frac{\xi_j^2}{2\gamma_j}. \quad (1.6)$$

All Wick-subtraction and  $a$  terms are affine in the last sum. A positive quadratic polynomial in that sum therefore dominates them at infinity. When  $\lambda = 0$ , the free harmonic operator supplies coercivity.

$\langle 2 \rangle 5$  This proves  $\langle 1 \rangle 2$ . □

$\langle 1 \rangle 3$  Construct the limiting sharp-time interaction in  $L^2(\mu)$ .

$\langle 2 \rangle 1$  Own-variance Wick ordering gives the exact chaos decomposition

$$V^{(K)} = \lambda I_4(f_K) + \lambda a I_2(g_K) + 2L\lambda b, \quad (1.7)$$

where  $f_K, g_K$  are restrictions of fixed infinite-mode kernels.

*Because:*  $:X^r:_{\mathbb{E}X^2}$  is the  $r$ th homogeneous Wiener chaos. The integral imposes spatial momentum conservation and introduces no lower-chaos terms because the Wick constant in (1.2) is exactly the variance in (0.5).

⟨2⟩ 2 For  $K' \geq K$ , the quartic tail satisfies

$$\|I_4(f_{K'}) - I_4(f_K)\|_2^2 = 4!(2L)^{-2} \sum_{\substack{k_1+\dots+k_4=0 \\ \max_i |k_i| \in (K, K']}} \prod_{i=1}^4 \frac{1}{2\gamma(k_i)}. \quad (1.8)$$

*Because:* The Wick isometry contracts only equal chaos orders. Expanding four Fourier sums and integrating  $x, y \in \mathbb{T}_L$  supplies two Kronecker constraints; one is eliminated, leaving (1.8), including the factor  $4!$ .

⟨2⟩ 3 Let  $g(k) = (m + |k|)^{-1}$ . Then  $(2\gamma(k))^{-1} \leq g(k)$  and

$$(g * g)(k) \leq CL \frac{1 + \log(2 + |k|/m)}{m + |k|}, \quad (g * g * g)(k) \leq CL^2 \frac{[1 + \log(2 + |k|/m)]^2}{m + |k|}. \quad (1.9)$$

*Because:*  $m + |k| \leq 2\sqrt{m^2 + k^2}$ . For the first convolution split the summation into  $|u| \leq |k|/2$ ,  $|k - u| \leq |k|/2$ , and the complement. In the first region the second denominator is at least  $(m + |k|)/2$  and  $\sum_{|u| \leq R} (m + |u|)^{-1} \leq CL[1 + \log(1 + R/m)]$ ; the second region is symmetric. In the complement both denominators are at least a fixed multiple of  $m + |k|$ , and the remaining square tail is  $O(L/(m + |k|))$ . Repeat the same three-region split after convolving the first estimate with  $g$ ; the partial sum of  $(1 + \log)/(m + |u|)$  contributes the second logarithm. These are finite sums on the grid  $(\pi/L)\mathbb{Z}$ , so no continuum-limit interchange occurs.

⟨2⟩ 4 The part of (1.8) with  $|k_1| > K$  is at most

$$CL^3 \sum_{|k_1| > K} \frac{[1 + \log(2 + |k_1|/m)]^2}{(m + |k_1|)^2} \leq CL^3 \frac{[1 + \log(1 + K/m)]^2}{K/m}. \quad (1.10)$$

*Because:* Sum the remaining three momenta by (1.9). Integral comparison of the decreasing tail  $\log^2 u/u^2$  gives the final bound. The union over the four choices of the large momentum changes only  $C$ .

⟨2⟩ 5 The quadratic tail has variance

$$\|I_2(g_{K'}) - I_2(g_K)\|_2^2 = \frac{1}{2} \sum_{K < |k| \leq K'} \gamma(k)^{-2} \leq \frac{CL}{K}. \quad (1.11)$$

*Because:* In the real sine/cosine basis it is the sum of independent multiples of  $\xi_j^2 - 1$ , whose variance is 2. The mode normalizations give the factor  $1/2$ ; compare the remaining sum with  $(2L/\pi) \int_K^\infty (m^2 + u^2)^{-1} du$ .

⟨2⟩ 6 There is a chaos- $\leq 4$  random variable  $V_L \in L^2(\mu)$  such that

$$\|V^{(K)} - V^{(K')}\|_2 + \|V^{(K)} - V_L\|_2 \leq C_V \frac{1 + \log(1 + K/m)}{(K/m)^{1/2}}. \quad (1.12)$$

*Because:* Combine (1.10), (1.11), orthogonality of distinct chaoses, and completeness of  $L^2(\mu)$ . The constant chaos cancels. The limiting kernels are precisely the truncation-convention objects denoted  $\lambda \int (\cdot; \varphi^4_{C_L} + a; \varphi^2_{C_L} + b) dx$ .

⟨2⟩ 7  $V^{(K)} \rightarrow V_L$  in every finite  $L^p(\mu)$ .

*Because:* If  $X$  is a sum of chaoses of orders at most four, (0.8) applied with  $A = e^{-\tau} \mathbf{1}$ ,  $q = 1 + e^{2\tau}$ , and the orthogonal chaos decomposition gives

$$\|X\|_q \leq (q - 1)^2 \|X\|_2 \quad (q \geq 2). \quad (1.13)$$

Use (1.13) for the difference and interpolate for  $p < 2$ .

⟨2⟩ 8 This proves ⟨1⟩3. □

### 3 The three Gaussian path ensembles and uniform exponential bounds

⟨1⟩ 4 Fix the three ensembles and their covariance identities.

⟨2⟩ 1 The vacuum ensemble is  $(Q, \mu)$  of (0.4).

⟨2⟩ 2 The periodic ensemble  $\mu_{t,L}^{\text{per}}$  is the centered field on  $\mathbb{T}_t \times \mathbb{T}_L$  with per-mode covariance

$$c_\omega(s - s') = \frac{\cosh(\omega(t/2 - |s - s'|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}, \quad \omega = \gamma(k). \quad (2.1)$$

⟨2⟩ 3 The stationary Ornstein–Uhlenbeck ensemble is the Gaussian process  $X_s = (\xi_j(s))_j$  with

$$\mathbb{E}[\xi_j(s)\xi_j(s')] = e^{-\gamma_j|s-s'|}. \quad (2.2)$$

For bounded cylinder  $F, G$ ,

$$\mathbb{E}[\overline{F(X_0)}G(X_s)] = \langle F, \Gamma(e^{-s\gamma})G \rangle_{L^2(\mu)}. \quad (2.3)$$

*Because:* Both sides of (2.3) are finite-dimensional Gaussian integrals with unit marginals and cross covariance  $e^{-s\gamma_j}$  in coordinate  $j$ . Equality for cylinders follows from their characteristic functions;  $L^2$  density extends it.

⟨2⟩ 4 This proves ⟨1⟩4. □

⟨1⟩ 5 Prove the abstract uniform Nelson lemma used at every cutoff.

Let  $A_K$  be chaos- $\leq 4$  random variables indexed by a full tail  $\mathcal{K} \subset \Gamma_\infty$ . Define  $\ell_K = 1 + \log(1 + K/m)$  and assume

$$\ell_{K^+} - \ell_K \leq \log 2, \quad (2.4)$$

$$A_K \geq -c_1 \ell_K^2, \quad (2.5)$$

$$\|A_K - A_{K'}\|_2 \leq C_A \ell_K (K/m)^{-\kappa} \quad (K' \geq K), \quad \kappa > 0. \quad (2.6)$$

⟨2⟩ 1 For  $b$  with  $L_b = \sqrt{b/(2c_1)}$  sufficiently large and  $b \leq c_1 \ell_K^2$ , choose the largest  $K_b \leq K$  for which  $\ell_{K_b} \leq L_b$ . Then

$$\frac{L_b}{2} < \ell_{K_b} \leq L_b. \quad (2.7)$$

*Because:* Maximality gives  $\ell_{K_b^+} > L_b$  and (2.4) gives  $\ell_{K_b} > L_b - \log 2 \geq L_b/2$ . If  $b > c_1 \ell_K^2$ , (2.5) makes  $\mathbb{P}(A_K \leq -b) = 0$ .

⟨2⟩ 2  $A_{K_b} \geq -b/2$ , hence

$$\{A_K \leq -b\} \subset \{A_K - A_{K_b} \leq -b/2\}. \quad (2.8)$$

⟨2⟩ 3 The  $L^2$  scale of the difference obeys

$$\rho_b := \|A_K - A_{K_b}\|_2 \leq 2C_A L_b e^{-\kappa(L_b/2-1)}. \quad (2.9)$$

*Because:*  $K_b/m = e^{\ell_{K_b}-1} - 1$ . For large  $L_b$ , this is at least  $\frac{1}{2}e^{\ell_{K_b}-1}$ ; insert (2.7) into (2.6).

⟨2⟩ 4 For every integer  $q \geq 1$ ,

$$\mathbb{P}(A_K \leq -b) \leq \left[ \frac{2(2q-1)^2 \rho_b}{b} \right]^{2q}. \quad (2.10)$$

*Because:* Apply Markov to  $|A_K - A_{K_b}|^{2q}$  in (2.8) and apply (1.13) with exponent  $2q$  to the chaos- $\leq 4$  difference.

⟨2⟩ 5 Choose  $q$  so that  $(2q)^2 \leq e^{\kappa L_b/4} < (2q+2)^2$ . After increasing the fixed threshold  $b_1$ ,

$$\mathbb{P}(A_K \leq -b) \leq \exp[-e^{\kappa L_b/8}] \quad (b \geq b_1), \quad (2.11)$$

uniformly in  $K$ .

*Because:* Substitute (2.9) into (2.10). The logarithm of the bracket is at most  $-\kappa L_b/8$  after the polynomial factors in  $L_b$  are absorbed;  $2q \geq \frac{1}{2}e^{\kappa L_b/8}$  gives (2.11), after another harmless increase of  $b_1$ .

⟨2⟩ 6 For every  $p < \infty$ ,

$$\sup_K \mathbb{E}[e^{-pA_K}] < \infty. \quad (2.12)$$

*Because:* Layer cake gives

$$\mathbb{E}[e^{-pA_K}] \leq e^{pb_1} + p \int_{b_1}^{\infty} e^{pb} \mathbb{P}(A_K \leq -b) db.$$

Since  $L_b$  is a positive constant times  $\sqrt{b}$ , the double-exponential decay in (2.11) dominates  $e^{pb}$  and makes the integral finite uniformly in  $K$ . If  $A_K \rightarrow A$  in  $L^2$ , take an a.s. convergent subsequence and apply Fatou to obtain the same bound for  $e^{-A}$ .

⟨2⟩ 7 The bounded successor-gap hypothesis (2.4) holds on every full lattice tail.

*Because:* For  $K^+ = K + \pi/L$  and  $K \geq \pi/L$ ,

$$\ell_{K^+} - \ell_K = \log \left( 1 + \frac{\pi/L}{m+K} \right) < \log 2.$$

This condition is essential to the scale selection (2.7); (2.5)–(2.6) alone do not imply that selection for a tower-sparse index set.

⟨2⟩ 8 This proves ⟨1⟩5. □

⟨1⟩ 6 Instantiate the Nelson lemma in all three ensembles.

⟨2⟩ 1 For  $0 < \sigma \leq \sigma_0$ ,

$$\sup_K \|e^{-\sigma V^{(K)}}\|_{L^p(\mu)} < \infty, \quad e^{-\sigma V_L} \in L^p(\mu). \quad (2.13)$$

*Because:* Conditions (2.5) and (2.6) are respectively (1.5) and (1.12), with  $\kappa = 1/2$ , after multiplying constants by  $\sigma_0$ .

⟨2⟩ 2 For

$$\mathcal{W}_K(t) = \int_0^t V^{(K)}(X_s) ds, \quad \mathcal{W}(t) = \int_0^t V_L(X_s) ds, \quad (2.14)$$

one has, uniformly for  $0 < t \leq T$ ,

$$\|\mathcal{W}_K - \mathcal{W}_{K'}\|_2 \leq t \|V^{(K)} - V^{(K')}\|_2, \quad \mathcal{W}_K \geq -TC\ell_K^2, \quad \sup_K \|e^{-\mathcal{W}_K}\|_p < \infty. \quad (2.15)$$

*Because:* Stationarity and Cauchy–Schwarz give

$$\|\mathcal{W}_K - \mathcal{W}_{K'}\|_2^2 = \int_0^t \int_0^t \mathbb{E}[\Delta V(X_s) \Delta V(X_u)] ds du \leq t^2 \|\Delta V\|_2^2.$$

The pathwise lower bound is the integral of (1.5). Time integration of a global Gaussian chaos remains in chaos orders at most four, so Step  $\langle 1 \rangle 5$  applies.

$\langle 2 \rangle 3$  Let  $\mathcal{U}_t^{(K)}$  be the spatial-mode truncation of the periodic interaction. Then

$$\mathcal{U}_t^{(K)} \rightarrow \mathcal{U}_t \quad \text{in } L^2(\mu_{t,L}^{\text{per}}), \quad \sup_K \|e^{-\mathcal{U}_t^{(K)}}\|_p < \infty. \quad (2.16)$$

$\langle 3 \rangle 1$  The pathwise lower bound is  $\mathcal{U}_t^{(K)} \geq -Ct\ell_K^2$ .

*Because:* Apply (1.4) at every spacetime point using the vacuum cutoff constant  $c_K$ , which obeys (1.1), and integrate over  $\mathbb{T}_t \times \mathbb{T}_L$ .

$\langle 3 \rangle 2$  The quadratic chaos tail satisfies

$$\|Y_2^{(K')} - Y_2^{(K)}\|_2^2 \leq C \sum_{|k| > K} \sum_{k_0 \in (2\pi/t)\mathbb{Z}} (m^2 + k_0^2 + k^2)^{-2} \leq C(t, L, m)K^{-2}. \quad (2.17)$$

*Because:* The first inequality is the Wick isometry and the continuum covariance envelope. Summing  $k_0$  by comparison with its integral gives  $\sum_{k_0} (A^2 + k_0^2)^{-2} \leq C(t_0)(A^{-4} + tA^{-3})$ ; the remaining one-dimensional spatial tail is  $O(tLK^{-2})$  (and the  $A^{-4}$  term is smaller under  $tL \geq 1$ ).

$\langle 3 \rangle 3$  The quartic chaos tail satisfies the explicit bound

$$\|Y_4^{(K')} - Y_4^{(K)}\|_2^2 \leq C(t, L, m) \sum_{\substack{|k| > K \\ k_0 \in (2\pi/t)\mathbb{Z}}} \frac{[1 + \log(2 + |(k_0, k)|)]^2}{(m^2 + k_0^2 + k^2)^2} \leq C(t, L, m)K^{-2}[1 + \log K]^2. \quad (2.18)$$

*Because:* Select one of the four momenta whose spatial component exceeds  $K$ ; there are four choices. Sum the other three constrained covariance factors using the two-dimensional convolution estimate already proved in the repaired V2 module:  $c_{\text{env}}^{*3}(-\kappa) \leq C(tL)^2[1 + \log(2 + |\kappa|)]^2/(m^2 + |\kappa|^2)$ . Multiplication by the selected factor gives the first sum. Cover the anisotropic grid by its fundamental rectangles. Its radial majorant is decreasing outside a fixed disk, so the sum is bounded by its density times

$$\int_{r > K} \frac{r[1 + \log(2 + r)]^2}{(m^2 + r^2)^2} dr \leq C \frac{[1 + \log K]^2}{m^2 + K^2}.$$

This is the missing “same tail counts” calculation in the source V3 document.

$\langle 3 \rangle 4$  The thermal recombination coefficient tail is exponentially small.

*Because:*

$$d^{(K)}(t) = \frac{1}{2L} \sum_{|k| \leq K} \frac{1}{\gamma(k)(e^{t\gamma(k)} - 1)},$$

and for  $t \geq t_0$  the omitted summand is bounded by  $Ce^{-t_0|k|}/(m + |k|)$ . Hence  $|d^{(K)} - d^{(K')}| \leq Ce^{-t_0K/2}$  after enlarging  $C$ .

$\langle 3 \rangle 5$  Equations (2.17)–(2.18), the finite Wick recombination identity, and boundedness of  $d^{(K)}$  imply

$$\|\mathcal{U}_t^{(K)} - \mathcal{U}_t^{(K')}\|_2 \leq C[1 + \log(1 + K/m)]^2(K/m)^{-1/2}. \quad (2.19)$$

*Because:* The actual chaos tails from (2.17)–(2.18) are at least as small as  $K^{-1} \log K$  after taking square roots. The displayed  $K^{-1/2} \log^2 K$  rate is a weaker common bound that also covers the coefficient cross terms. Each cross term is a bounded coefficient difference times a bounded lower-chaos norm.

⟨3⟩ 6 Absorb one logarithm by  $\ell_K \leq C_\varsigma(K/m)^\varsigma$ , choose  $0 < \varsigma < 1/2$ , and apply Step ⟨1⟩5 with  $\kappa = 1/2 - \varsigma$ . This proves (2.16).

⟨2⟩ 4 The deterministically shifted periodic interactions obey the same Cauchy and exponential-integrability conclusions, uniformly for the bounded coarse-data class used in V4.

*Because:* Substitute  $u = \eta^{(K)}(s, x) + H^{(K)}(s, x)$  in (1.4). Since (1.4) holds for every real  $u$ , integration gives the same lower bound  $\mathcal{U}_t^{(K)}(\eta + H^{(K)}) \geq -Ct\ell_K^2$ . Expand each shifted Wick monomial by the Appell identity (3.12) and its degree-two analogue. The repaired V2 theorem, whose fundamental-band and alias sums are displayed in its Steps ⟨1⟩7–⟨1⟩8, bounds the resulting random-chaos tails and deterministic coefficient differences by the right side of (2.19), uniformly over the stated coarse-data bounds. These are precisely the lower-tail and  $L^2$ -increment hypotheses of Step ⟨1⟩5; applying that step proves the asserted uniform exponential integrability and  $L^2$  convergence.

⟨2⟩ 5 This proves ⟨1⟩6. □

⟨1⟩ 7 Derive uniform spectral lower bounds for the finite-cutoff operators.

Let  $K^{(K)}$  be defined in the next section. Its finite-mode periodic trace identity will be proved independently of this step.

⟨2⟩ 1 For a fixed admissible  $t_0$ ,

$$e^{-t_0 \inf \text{spec } K^{(K)}} \leq \text{Tr } e^{-t_0 K^{(K)}} = Z_0(t_0) \mathbb{E}[e^{-\mathcal{U}_{t_0}^{(K)}}] \leq Z_0(t_0) B. \quad (2.20)$$

*Because:* The first inequality selects the lowest heat eigenvalue; the identity is proved in Step ⟨1⟩8 below; the last inequality is (2.16) at  $p = 1$ . No uniform lower bound is used in proving that finite-mode identity, so this is not circular.

⟨2⟩ 2 Taking logarithms gives a  $K$ -independent constant  $c_\lambda$  with

$$K^{(K)} \geq -c_\lambda. \quad (2.21)$$

⟨2⟩ 3 For  $0 < \delta < 1$ , apply (2.21) to the amplified coupling  $\lambda/(1 - \delta)$  to obtain

$$K^{(K)} = \delta H_0 + (1 - \delta) [H_0 + (1 - \delta)^{-1} V^{(K)}] \geq \delta H_0 - c_\delta \quad (2.22)$$

as quadratic forms.

⟨2⟩ 4 This proves ⟨1⟩7. □

## 4 Finite-mode systems and their exact Feynman–Kac formulas

For fixed  $K$ , decompose  $L^2(Q, \mu) = L^2(\mu_{\leq K}) \otimes L^2(\mu_{> K})$  and set

$$K^{(K)} = (H_0^{\leq K} + V^{(K)}) \otimes \mathbf{1} + \mathbf{1} \otimes H_0^{> K}, \quad T_K(s) = e^{-sK^{(K)}}. \quad (3.1)$$

⟨1⟩ 8 Construct each finite-mode system and prove every representation used later.

⟨2⟩ 1 The form of  $H_0^{\leq K} + V^{(K)}$  is densely defined, lower bounded, and closed.

*Because:* Choose  $C > b_K$  and write

$$q_K(f) = \sum_{|k_j| \leq K} \gamma_j \|\partial_j f\|_2^2 + \int (V^{(K)} + C) |f|^2 d\mu_{\leq K}. \quad (3.2)$$

Both summands are nonnegative and their domains contain  $C_c^\infty(\mathbb{R}^{n_K})$ . If  $f_n$  is Cauchy in the norm  $\|f\|_2^2 + q_K(f)$ , it converges in  $L^2$  and in each weighted weak derivative. A subsequence converges pointwise a.e.; Fatou for the multiplication term and closedness of weak derivatives identify the limits. Thus the form norm is complete.

⟨2⟩ 2 The finite-mode operator has compact resolvent.

*Because:* On every Euclidean ball the Gaussian density is bounded above and below by positive constants, so the weighted form norm controls the ordinary  $H^1$  norm there; Rellich makes the restricted embedding compact. Outside a sufficiently large ball, coercivity from (1.6) gives  $(V^{(K)} + C) \geq R$ , hence  $\int_{|\xi| > R_0} |f|^2 d\mu \leq R^{-1} q_K(f)$ . First take  $R$  large and then use Rellich on the ball. The unit ball of the form domain is therefore relatively compact in  $L^2$ .

⟨2⟩ 3 For one retained coordinate, the free Mehler kernel with respect to  $\mathcal{N}(0, 1)$  is

$$M_s^{(j)}(x, y) = (1 - r_j^2)^{-1/2} \exp \left[ -\frac{r_j^2(x^2 + y^2) - 2r_j xy}{2(1 - r_j^2)} \right], \quad r_j = e^{-s\gamma_j}. \quad (3.3)$$

*Because:* Completing the square shows  $\int M_s^{(j)}(x, y) d\mu(y) = 1$ ; symmetry is evident. Applying the Gaussian generating function to Hermite polynomials gives eigenvalue  $r_j^n$  in degree  $n$ . Hence the product over retained coordinates is exactly  $\Gamma(e^{-s\gamma})$  on that finite Gaussian space.

⟨2⟩ 4 For bounded  $f, g$  on the retained coordinates,

$$\langle f, e^{-s(H_0^{\leq K} + V^{(K)})} g \rangle = \mathbb{E} \left[ \overline{f(X_0^{\leq K})} g(X_s^{\leq K}) e^{-\int_0^s V^{(K)}(X_u) du} \right]. \quad (3.4)$$

⟨3⟩ 1 First replace  $V^{(K)}$  by  $V_R = V^{(K)} \wedge R$ , with  $R > 0$ .

*Because:* By (1.5),  $-b_K \leq V_R \leq R$ , so multiplication by  $V_R$  is bounded and self-adjoint.

⟨3⟩ 2 The bounded-perturbation Trotter formula gives

$$e^{-s(H_0 + V_R)} = \lim_{n \rightarrow \infty} (e^{-(s/n)H_0} e^{-(s/n)V_R})^n. \quad (3.5)$$

⟨3⟩ 3 Insertion of (3.3) between the multiplication factors turns the matrix element of the product into the OU chain integral with weight

$$\exp \left[ -\frac{s}{n} \sum_{j=1}^n V_R(X_{js/n}) \right].$$

⟨3⟩ 4 As  $n \rightarrow \infty$  this weight converges a.s. to  $e^{-\int_0^s V_R(X_u) du}$  and is bounded by  $e^{sb_K}$ .

*Because:* The finite-dimensional OU process has continuous paths and  $V_R$  is bounded continuous. The displayed sum is therefore an ordinary Riemann sum; dominated convergence applies.

⟨3⟩ 5 As  $R \rightarrow \infty$ , the operators and path integrals converge to (3.4).

*Because:*  $V_R \uparrow V^{(K)}$ , so the associated closed forms increase and monotone form convergence gives strong semigroup convergence. On each continuous path,  $e^{-\int V_R} \downarrow e^{-\int V^{(K)}}$ ; it is bounded by  $e^{sb_K}$ , so dominated convergence applies. This proves (3.4).

⟨2⟩ 5 The kernel in (3.4) is jointly continuous and strictly positive.

*Because:* Condition the OU law on its two endpoints. The product Mehler density is continuous and strictly positive. The bridge expectation of the weight lies in  $(0, e^{sb_K}]$ . Continuity follows by dominated convergence after realizing the finite-dimensional Gaussian bridge as an affine continuous function of its endpoints plus an endpoint-independent bridge.

⟨2⟩ 6 Tensoring with the independent free tail gives, for bounded cylinders  $f, g$ ,

$$\langle f, T_K(s)g \rangle = \mathbb{E} \left[ \overline{f(X_0)} g(X_s) e^{-W_K(s)} \right]. \quad (3.6)$$

*Because:* The retained and omitted OU coordinates are independent; apply (3.4) to the first factor and (2.3) to the second.

⟨2⟩ 7  $T_K(s)$  is trace class and

$$\text{Tr } T_K(s) \leq e^{sb_K} Z_0(s). \quad (3.7)$$

*Because:* The min–max principle and  $V^{(K)} \geq -b_K$  imply that every ordered eigenvalue of the retained interacting operator is at least the corresponding free eigenvalue minus  $b_K$ . Tensoring with the free tail and summing all occupation numbers gives (3.7). In particular, the full tensor operator has compact resolvent.

⟨2⟩ 8 The periodic trace formula is

$$\text{Tr } T_K(s) = Z_0(s) \mathbb{E}_{\mu_{s,L}^{\text{per}}} [e^{-\mathcal{U}_s^{(K)}}]. \quad (3.8)$$

*Because:* Apply the diagonal trace to the strictly positive kernel from (3.4), normalize the product of cyclic Mehler kernels by its free trace, and integrate the intermediate points. The normalized cyclic Gaussian chain has covariance (2.1): for one mode, direct inversion of the cyclic tridiagonal quadratic form gives  $\cosh(\omega(s/2 - |u - v|)) / (2\omega \sinh(\omega s/2))$ . The equal-time variance differs from  $c_K$  by  $d^{(K)}(s)$ ; the exact Wick-recombination identity

$$:X^4_{\cdot c_K} = :X^4_{\cdot v_K} + 6(v_K - c_K):X^2_{\cdot v_K} + 3(v_K - c_K)^2 \quad (3.9)$$

identifies the path action with  $\mathcal{U}_s^{(K)}$ . Passing from equal cyclic partitions to the continuous bridge uses the bounded-truncation and dominated limits already justified in Steps ⟨3⟩1–⟨3⟩5.

⟨2⟩ 9 For a Weyl symbol whose position and momentum components are supported in  $|k| \leq K$ ,

$$\frac{\text{Tr}(T_K(s)W_{\text{ct}}(\xi))}{\text{Tr } T_K(s)} = e^{-S^{(K)}(H_p)} \frac{\mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_s^{(K)}(\eta + H_p^{(K)})}]}{\mathbb{E}[e^{-\mathcal{U}_s^{(K)}}]}. \quad (3.10)$$

Here

$$S^{(K)}(H_p) = \frac{1}{8L} \sum_{|k| \leq K} \gamma(k) \coth(s\gamma(k)/2) |b_\infty(k)|^2. \quad (3.11)$$

*Because:* Apply the V1 proof mode by mode: the Weyl operator translates the endpoint by its momentum component and multiplies by its position phase; the unique solution of  $(-\partial_u^2 + \gamma(k)^2)H = 0$  off the marked slice with the prescribed jump removes the endpoint translation; completing the Gaussian square produces (3.11) and the sharp-time phase. The Wick Appell identity

$$:(X + H)^4_{\cdot c} = :X^4_{\cdot c} + 4H:X^3_{\cdot c} + 6H^2:X^2_{\cdot c} + 4H^3X + H^4 \quad (3.12)$$

gives the shifted action without an extra  $-6c(2HX + H^2)$  term. To remove the interaction cutoff, replace  $V^{(K)}$  by  $V_R = V^{(K)} \wedge R$  exactly as in Steps  $\langle 3 \rangle 1$ – $\langle 3 \rangle 5$ : the Weyl insertion is unitary, the truncated path weight is bounded by  $e^{sb_K}$ , and dominated convergence first passes the Trotter mesh and then  $R \rightarrow \infty$ . All retained spaces are finite-dimensional. Tensoring with the omitted free modes produces the same free factor in numerator and denominator, so that factor cancels.

$\langle 2 \rangle 10$  The kernel is positivity improving, so the finite-mode lowest eigenvalue is simple.

*Because:* If  $f, g \geq 0$  are nonzero, the endpoint Mehler density is strictly positive and the Feynman–Kac weight is strictly positive on every finite path; therefore  $\langle f, e^{-s(H_0^{\leq K} + V^{(K)})} g \rangle > 0$ . Let  $T$  denote this compact self-adjoint integral operator and let  $Th = \|T\|h$ . Its strictly positive kernel gives  $|Th(x)| \leq T|h|(x)$ ; the variational bound  $\langle |h|, T|h| \rangle \leq \|T\| \|h\|^2$  forces equality. Equality in the integral triangle inequality for a strictly positive kernel forces  $h(y)$  to have one common phase for almost every  $y$ . Thus every top eigenvector is a scalar multiple of one strictly positive vector, so the top eigenvalue is simple. Tensoring with the free tail preserves a unique total ground vector because each strictly positive tail frequency has the unique oscillator vacuum.

$\langle 2 \rangle 11$  This proves  $\langle 1 \rangle 8$  and supplies the identity used in (2.20).  $\square$

## 5 Strong semigroup limit and the continuum generator

$\langle 1 \rangle 9$  Prove that  $T_K(s)$  converges strongly to one self-adjoint semigroup.

$\langle 2 \rangle 1$  For a bounded cylinder  $\psi$  supported in modes at most  $K_0$  and  $K, K' \geq K_0$ ,

$$\|(T_K(s) - T_{K'}(s))\psi\|_2 \leq \|\psi\|_\infty \|e^{-\mathcal{W}_K(s)} - e^{-\mathcal{W}_{K'}(s)}\|_2. \quad (4.1)$$

*Because:* Subtract the two conditional-expectation versions of (3.6), condition on  $X_0$ , square, integrate, and apply conditional Jensen.

$\langle 2 \rangle 2$  For real  $x, y$ ,

$$|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y}). \quad (4.2)$$

*Because:* Integrate the derivative of  $e^{-[(1-u)x+uy]}$  over  $u \in [0, 1]$  and bound every intermediate exponential by  $e^{-x} + e^{-y}$ .

$\langle 2 \rangle 3$  Equations (1.13), (2.13)–(2.15), and Hölder give

$$\|e^{-\mathcal{W}_K} - e^{-\mathcal{W}_{K'}}\|_2 \leq C_T s \|V^{(K)} - V^{(K')}\|_2. \quad (4.3)$$

*Because:* Apply (4.2), then  $\|XY\|_2 \leq \|X\|_4 \|Y\|_4$  with  $X = \mathcal{W}_K - \mathcal{W}_{K'}$  and  $Y = e^{-\mathcal{W}_K} + e^{-\mathcal{W}_{K'}}$ . The first norm is at most  $9s\|V^{(K)} - V^{(K')}\|_2$  by (1.13), (2.15); the second is uniformly bounded by (2.13)–(2.15).

$\langle 2 \rangle 4$   $T_K(s)$  is strongly Cauchy on  $L^2(\mu)$ , uniformly for  $s$  in a compact interval.

*Because:* Equations (4.1), (4.3), and (1.12) prove this on the dense bounded cylinder core. The uniform operator bound  $\|T_K(s)\| \leq e^{sc_\lambda}$  from (2.21) extends it to all  $L^2$ . Call the strong limit  $T(s)$ .

$\langle 2 \rangle 5$   $T(s)$  is a self-adjoint semigroup with  $\|T(s)\| \leq e^{sc_\lambda}$ .

*Because:* Strong limits preserve the adjoint identity on matrix elements. For the product, insert and subtract  $T_K(s)T_K(u)$  and use uniform boundedness to pass both factors strongly:  $T(s)T(u) = T(s+u)$ . The norm bound passes by lower semicontinuity.

⟨2⟩ 6  $T(s)$  is strongly continuous at  $s = 0$ .

*Because:* For a bounded cylinder  $\psi$  and fixed  $K$  containing its modes,

$$\|T(s)\psi - \psi\| \leq \|T(s)\psi - T_K(s)\psi\| + \|T_K(s)\psi - \psi\|.$$

By (4.1)–(4.3), the first term is at most  $Cs\|\psi\|_\infty \sup_{K' \geq K} \|V^{(K')} - V^{(K)}\|_2$ ; the second tends to zero for fixed  $K$ . First let  $s \downarrow 0$ , then  $K \rightarrow \infty$ . Density and the uniform norm bound finish.

⟨2⟩ 7 There is a unique self-adjoint  $K_L \geq -c_\lambda$  such that

$$T(s) = e^{-sK_L}. \quad (4.4)$$

*Because:* Apply the spectral-generator theorem for a strongly continuous self-adjoint semigroup bounded by  $e^{sc_\lambda}$ . The semigroup uniquely determines its generator.

⟨2⟩ 8 For bounded measurable  $f, g$ ,

$$\langle f, T(s)g \rangle = \mathbb{E}[\overline{f(X_0)}g(X_s)e^{-\mathcal{W}(s)}]. \quad (4.5)$$

*Because:* For bounded cylinders, pass  $K \rightarrow \infty$  in (3.6) by (4.3) and strong convergence. Approximate bounded measurable functions by their conditional expectations on the increasing mode sigma-fields. Martingale convergence gives a.e. and  $L^2$  convergence; dominated convergence on the right uses  $e^{-\mathcal{W}} \in L^1$ , while strong continuity of  $T(s)$  handles the left.

⟨2⟩ 9 The uniform form bound passes to the limit:

$$K_L \geq \tfrac{1}{2}H_0 - c_{1/2}. \quad (4.6)$$

⟨3⟩ 1 Choose  $M > c_{1/2} \vee c_\lambda$ . From (2.22),

$$(K^{(K)} + M)^{-1} \leq (\tfrac{1}{2}H_0 + M - c_{1/2})^{-1}. \quad (4.7)$$

*Because:* For strictly positive self-adjoint  $S$ , spectral calculus gives

$$q_S(\psi) = \sup_{\phi} [2 \operatorname{Re} \langle \psi, \phi \rangle - ip\phi S^{-1}\phi], \quad \langle \phi, S^{-1}\phi \rangle = \sup_{\psi} [2 \operatorname{Re} \langle \psi, \phi \rangle - q_S(\psi)]. \quad (4.8)$$

Thus form order and the reverse resolvent order are equivalent.

⟨3⟩ 2 The resolvents converge strongly:

$$(K^{(K)} + M)^{-1} o(K_L + M)^{-1}. \quad (4.9)$$

*Because:* Use the Laplace representation  $S^{-1} = \int_0^\infty e^{-Ms} e^{-s(S-M)} ds$  with the semigroups  $T_K, T$ ;  $M > c_\lambda$  supplies an integrable uniform majorant.

⟨3⟩ 3 Pass to the limit in (4.7) on every vector and use (4.8) in the reverse direction. This gives (4.6) and  $Q(K_L) \subset Q(H_0)$ .

⟨2⟩ 10 This proves ⟨1⟩ 9. □

⟨1⟩ 10 Identify the generator and its form on the cylinder core.

⟨2⟩ 1 For  $\psi \in \mathcal{D}$ ,

$$\frac{(1 - T(s))\psi}{s} = \frac{(1 - e^{-sH_0})\psi}{s} - \frac{1}{s} \mathbb{E}[\psi(X_s)(e^{-\mathcal{W}(s)} - 1) \mid X_0]. \quad (4.10)$$

*Because:* This is (4.5) written as a conditional expectation, with the free conditional expectation subtracted and added.

⟨2⟩ 2 The first term in (4.10) converges in  $L^2$  to  $H_0\psi$ .

*Because:* Let  $d\mu_\psi(\lambda) = d\langle\psi, E_{H_0}(\lambda)\psi\rangle$ . The spectral theorem gives

$$\left\| \frac{1 - e^{-sH_0}}{s} \psi - H_0\psi \right\|_2^2 = \int_0^\infty \left| \frac{1 - e^{-s\lambda}}{s} - \lambda \right|^2 d\mu_\psi(\lambda).$$

The integrand converges pointwise to zero and is bounded by  $\lambda^2$ , because  $0 \leq (1 - e^{-s\lambda})/s \leq \lambda$ . Since  $\psi \in D(H_0)$ ,  $\int \lambda^2 d\mu_\psi < \infty$ ; dominated convergence proves the claim.

⟨2⟩ 3 Write  $e^{-x} - 1 = -x + \rho(x)$ , where

$$|\rho(x)| \leq \frac{1}{2}x^2 e^{x-}, \quad x_- = \max(-x, 0). \quad (4.11)$$

*Because:* Taylor's formula with integral remainder gives  $\rho(x) = x^2 \int_0^1 (1-u)e^{-ux} du$ ; bound  $e^{-ux} \leq e^{x-}$ .

⟨2⟩ 4 The linear part satisfies

$$\frac{1}{s} \mathbb{E}[\psi(X_s) \mathcal{W}(s) \mid X_0] = \frac{1}{s} \int_0^s e^{-uH_0} (V_L e^{-(s-u)H_0} \psi) du \longrightarrow V_L \psi \quad (4.12)$$

in  $L^2$ .

*Because:* The Markov property gives the identity. Since  $V_L \in L^4$  and  $\psi$  is bounded,

$$\|V_L(e^{-vH_0}\psi - \psi)\|_2 \leq \|V_L\|_4 \|e^{-vH_0}\psi - \psi\|_4.$$

For  $\psi = F(\xi_1, \dots, \xi_n)$ , the Mehler formula and bounded derivatives of  $F$  give

$$\sup_{0 \leq v \leq s} \|e^{-vH_0}\psi - \psi\|_4 \leq C_F \sum_{j \leq n} (1 - e^{-s\gamma_j})^{1/2} \rightarrow 0.$$

Also  $e^{-uH_0}$  is strongly continuous on  $V_L\psi \in L^2$ . Both errors are uniform for  $0 \leq u \leq s$ , proving (4.12).

⟨2⟩ 5 The remainder in (4.11), divided by  $s$ , tends to zero in  $L^2$ .

*Because:* Conditional Jensen, Hölder, (1.13), and (2.15) give

$$\frac{1}{s} \|\mathbb{E}[\psi(X_s) \rho(\mathcal{W}) \mid X_0]\|_2 \leq \frac{\|\psi\|_\infty}{2s} \|\mathcal{W}\|_8^2 \|e^{\mathcal{W}-}\|_4 \leq Cs. \quad (4.13)$$

Indeed  $\|\mathcal{W}\|_8 \leq 49s\|V_L\|_2$  and  $e^{\mathcal{W}-} \leq 1 + e^{-\mathcal{W}}$ , whose  $L^4$  norm is uniformly bounded by Step ⟨1⟩6.

⟨2⟩ 6 Therefore

$$\mathcal{D} \subset D(K_L), \quad K_L\psi = H_0\psi + V_L\psi \quad (\psi \in \mathcal{D}). \quad (4.14)$$

*Because:* Combine (4.10)–(4.13) with the generator characterization  $K_L\psi = \lim_{s \downarrow 0} s^{-1}(1 - e^{-sK_L})\psi$ .

⟨2⟩ 7 For  $\psi \in \mathcal{D}$ ,

$$q_{K_L}(\psi) = \langle \psi, H_0\psi \rangle + \int V_L |\psi|^2 d\mu. \quad (4.15)$$

*Because:* Spectral calculus gives  $s^{-1} \langle \psi, (1 - e^{-sK_L})\psi \rangle \rightarrow q_{K_L}(\psi)$  after shifting by the lower spectral bound. Equation (4.14) identifies the same limit with the right side. This proves equality on  $\mathcal{D}$  only; no unproved claim that  $\mathcal{D}$  is a form core is made.

⟨2⟩ 8 This proves ⟨1⟩10. □

## 6 Uniform smoothing, trace class, and trace-norm approximation

⟨1⟩ 11 Prove the cutoff-uniform  $L^2 \rightarrow L^4$  smoothing estimate.

Put  $s_4 = (\log 3)/m$ .

⟨2⟩ 1 By self-adjoint duality it suffices to prove

$$\sup_K \|T_K(s_4)\|_{4/3 \rightarrow 2} < \infty. \quad (5.1)$$

*Because:*  $\|T_K(s_4)\|_{2 \rightarrow 4} = \|T_K(s_4)\|_{4/3 \rightarrow 2}$ . For  $s > s_4$ , compose with  $\|T_K(s - s_4)\|_{2 \rightarrow 2} \leq e^{c\lambda(s-s_4)}$ .

⟨2⟩ 2 Replace  $W = V^{(K)}$  by  $W_R = W \wedge R$  and use Trotter with  $\tau = s_4/n$ . It is enough to bound

$$(e^{-\tau W_R} e^{-\tau H_0})^n : L^{4/3} \rightarrow L^2 \quad (5.2)$$

uniformly in  $n, K, R$ .

*Because:* The bounded Trotter products converge strongly. The form semigroups for  $W_R \uparrow W$  converge strongly. An a.e. convergent subsequence and Fatou pass any common  $L^2$  bound through both limits.

⟨2⟩ 3 For  $j = 0, \dots, n$ , define

$$p_j = 1 + \frac{1}{3}e^{j\tau m}, \quad P_j = 1 + (p_j - 1)e^{2\tau m}. \quad (5.3)$$

Then  $p_0 = 4/3$ ,  $p_n = 2$ , and

$$\|e^{-\tau H_0}\|_{p_j \rightarrow P_j} \leq 1. \quad (5.4)$$

*Because:*  $\|e^{-\tau \gamma}\| = e^{-\tau m}$  and  $e^{-\tau m} = \sqrt{(p_j - 1)/(P_j - 1)}$ ; apply (0.8).

⟨2⟩ 4 Define  $r_j$  by

$$\frac{1}{p_{j+1}} = \frac{1}{r_j} + \frac{1}{P_j}. \quad (5.5)$$

Then multiplication gives

$$\|e^{-\tau W_R} u\|_{p_{j+1}} \leq \|e^{-\tau W_R}\|_{r_j} \|u\|_{P_j}. \quad (5.6)$$

⟨2⟩ 5 For  $n \geq 2$ ,

$$\sum_{j=0}^{n-1} \frac{1}{r_j} \leq 1, \quad r_j \tau \leq \frac{12}{m}. \quad (5.7)$$

*Because:* From (5.3),  $P_j - p_{j+1} = \frac{1}{3}e^{(j+1)\tau m}(e^{\tau m} - 1)$ . Since  $p_{j+1}, P_j \geq 4/3$ ,

$$\sum_j r_j^{-1} \leq \frac{3}{16}e^{\tau m}(e^{n\tau m} - 1) = \frac{3}{8}e^{\tau m} \leq \frac{3\sqrt{3}}{8} < 1.$$

Also  $p_{j+1} \leq 2$ ,  $P_j \leq 1 + \frac{1}{3}e^{(j+2)\tau m}$ , and  $\tau m \leq e^{\tau m} - 1$ , whence

$$r_j \tau m = \frac{3\tau m, p_{j+1} P_j}{e^{(j+1)\tau m}(e^{\tau m} - 1)} \leq 6 + 2e^{\tau m} < 12.$$

⟨2⟩ 6 There is a constant  $D$  independent of  $j, n, K, R$  such that

$$\|e^{-\tau W_R}\|_{r_j} \leq D^{1/r_j}. \quad (5.8)$$

*Because:* Since  $R > 0$ ,  $e^{-r_j \tau W_R} \leq 1 + e^{-r_j \tau W}$ . Equation (5.7) restricts  $r_j \tau$  to  $(0, 12/m]$ , so the vacuum Nelson bound (2.13), with  $p = 1$  and all  $\sigma$  in that compact interval, makes  $\mathbb{E}e^{-r_j \tau W}$  uniformly bounded.

⟨2⟩ 7 Multiplication of (5.4), (5.6), and (5.8) yields

$$\|(e^{-\tau W_R} e^{-\tau H_0})^n\|_{4/3 \rightarrow 2} \leq D \sum_j 1/r_j \leq \max(1, D) =: M_0. \quad (5.9)$$

⟨2⟩ 8 Consequently, uniformly in  $K$ ,

$$\|T_K(s)\|_{2 \rightarrow 4} \leq M_0 e^{c_\lambda(s-s_4)} \quad (s \geq s_4), \quad (5.10)$$

and the same holds for  $T(s)$  by strong convergence and Fatou.

⟨2⟩ 9 This proves ⟨1⟩ 11. □

⟨1⟩ 12 Upgrade strong convergence to operator-norm convergence at one time.

Set  $s_2 = 2s_4$  and  $\Delta V = V^{(K')} - V^{(K)}$ .

⟨2⟩ 1 For bounded cylinder  $\phi, \psi$  and every  $0 < s < s_2$ , both  $T_K(s)\phi$  and  $T_{K'}(s_2 - s)\psi$  lie in the form domains of both  $K^{(K)}$  and  $K^{(K')}$ .

*Because:* The uniform form lower bound (2.22) puts both vectors in  $Q(H_0)$ . From (3.6), conditional Jensen, and the OU Nelson bound,

$$\|T_K(u)\phi\|_4 \leq C\|\phi\|_\infty.$$

For every  $v \in L^4$  and either cutoff interaction,  $\int |V^{(\cdot)}|v|^2 d\mu \leq \|V^{(\cdot)}\|_2 \|v\|_4^2 < \infty$ . Thus both vectors also lie in both multiplication-form domains.

⟨2⟩ 2 The form Duhamel identity is

$$\langle \phi, (T_K(s_2) - T_{K'}(s_2))\psi \rangle = \int_0^{s_2} \langle T_K(u)\phi, \Delta V, T_{K'}(s_2 - u)\psi \rangle du. \quad (5.11)$$

*Because:* Differentiate  $u \mapsto \langle T_K(u)\phi, T_{K'}(s_2 - u)\psi \rangle$  in the common form domain. The two  $H_0$  form terms cancel, leaving multiplication by  $\Delta V$ . Integrability follows from the estimates in the next step, so the fundamental theorem for absolutely continuous form matrix elements applies.

⟨2⟩ 3 For  $0 \leq u \leq s_4$ , use exponents  $(2, 4, 4)$  on  $(T_K(u)\phi, \Delta V, T_{K'}(s_2 - u)\psi)$ ; for  $s_4 \leq u \leq s_2$ , reverse the two semigroup roles. Then

$$\|T_K(s_2) - T_{K'}(s_2)\|_{2 \rightarrow 2} \leq 18s_2 M_0 e^{2c_\lambda s_2} \|V^{(K)} - V^{(K')}\|_2. \quad (5.12)$$

*Because:* The long semigroup factor has elapsed time at least  $s_4$  and is  $L^2 \rightarrow L^4$  bounded by (5.10); the short factor is  $L^2 \rightarrow L^2$  bounded. The interaction difference satisfies  $\|\Delta V\|_4 \leq 9\|\Delta V\|_2$  by (1.13). Integrate the common bound over a length  $s_2$  interval; the harmless factor 2 accommodates both halves.

⟨2⟩ 4 Equation (1.12) implies

$$\|T_K(s_2) - T(s_2)\|_{2 \rightarrow 2} \rightarrow 0. \quad (5.13)$$

⟨2⟩ 5 This proves ⟨1⟩ 12. □

⟨1⟩ 13 Prove discreteness, eigenvalue convergence, trace class, and trace-norm convergence for every positive time.

Let  $\nu_0 \leq \nu_1 \leq \dots$  be the occupation-number eigenvalues of  $H_0$ .

$\langle 2 \rangle 1$   $T(s_2)$  is compact, and if  $\mu_j^{(K)}$  and  $\mu_j$  are ordered eigenvalues of  $K^{(K)}$  and  $K_L$ , then

$$\mu_j^{(K)} \rightarrow \mu_j, \quad \mu_j \geq \frac{1}{2}\nu_j - c_{1/2}. \quad (5.14)$$

*Because:* Each  $T_K(s_2)$  is trace class by (3.7), hence compact. The operator-norm limit (5.13) is compact. The min–max/singular-value inequality gives convergence of every ordered heat eigenvalue  $e^{-s_2\mu_j^{(K)}} \rightarrow e^{-s_2\mu_j}$ ; the logarithm is continuous on  $(0, \infty)$ . Apply min–max to (2.22) at  $\delta = 1/2$  and pass to the limit.

$\langle 2 \rangle 2$  For every  $s > 0$ ,

$$\mathrm{Tr} T_K(s) \rightarrow \mathrm{Tr} T(s) < \infty. \quad (5.15)$$

*Because:* By (5.14), each summand converges, while

$$e^{-s\mu_j^{(K)}} \leq e^{sc_{1/2}} e^{-s\nu_j/2}, \quad \sum_j e^{-s\nu_j/2} = Z_0(s/2) < \infty.$$

Dominated convergence on the counting measure gives (5.15), and shows that the spectrum of  $K_L$  is discrete and exhausts the  $\mu_j$ .

$\langle 2 \rangle 3$  For  $u = s/2$ ,

$$\|T_K(u) - T(u)\|_{\mathcal{J}_2}^2 = \mathrm{Tr} T_K(2u) - 2 \mathrm{Tr}(T_K(u)T(u)) + \mathrm{Tr} T(2u). \quad (5.16)$$

*Because:* Expand the square of the self-adjoint difference and use cyclicity for products of a trace-class and a bounded operator.

$\langle 2 \rangle 4$  The mixed trace in (5.16) converges to  $\mathrm{Tr} T(2u)$ .

*Because:* If  $(\phi_n)$  diagonalizes  $T(u)$  with eigenvalues  $\sigma_n$ , then

$$\mathrm{Tr}(T_K(u)T(u)) = \sum_n \sigma_n \langle \phi_n, T_K(u)\phi_n \rangle.$$

Each matrix element converges strongly, is bounded by  $e^{uc_\lambda}$ , and  $\sum_n \sigma_n = \mathrm{Tr} T(u) < \infty$ ; dominated convergence applies.

$\langle 2 \rangle 5$  Thus  $\|T_K(u) - T(u)\|_{\mathcal{J}_2} \rightarrow 0$ , and

$$\|T_K(s) - T(s)\|_{\mathcal{J}_1} \leq (\|T_K(u)\|_{\mathcal{J}_2} + \|T(u)\|_{\mathcal{J}_2}) \|T_K(u) - T(u)\|_{\mathcal{J}_2} \rightarrow 0. \quad (5.17)$$

*Because:* Factor  $T_K(s) - T(s) = T_K(u)[T_K(u) - T(u)] + [T_K(u) - T(u)]T(u)$  and apply the Schatten Hölder inequality. The Hilbert–Schmidt norms are bounded by (5.15).

$\langle 2 \rangle 6$  For every bounded  $B$ ,

$$\mathrm{Tr}(T_K(s)B) \rightarrow \mathrm{Tr}(T(s)B). \quad (5.18)$$

*Because:* The absolute difference is at most  $\|T_K(s) - T(s)\|_{\mathcal{J}_1} \|B\|$ .

$\langle 2 \rangle 7$  This proves  $\langle 1 \rangle 13$ . □

## 7 Periodic Feynman–Kac clause and Weyl insertions

Fix a coarse Weyl symbol  $\xi = \varepsilon_M q + ip$  and its continuum refinement. Let  $H_p$  be the V2 continuum harmonic interpolant,  $H_p^{(K)}$  its mode truncation, and let  $J_q, J_q^{(K)}$  be the corresponding sharp-time sources.

⟨1⟩ 14 Pass every deterministic shift datum to the continuum cutoff limit.

⟨2⟩ 1 The action tail satisfies

$$0 \leq S_\infty(H_p) - S^{(K)}(H_p) \leq C_\xi (K/m)^{-\eta}. \quad (6.1)$$

*Because:* The tail is

$$\frac{1}{8L} \sum_{|k| > K} \gamma(k) \coth(t\gamma(k)/2) |b_\infty(k)|^2.$$

The D4 envelope is  $|b_\infty(k)|^2 \leq C(1 + \varepsilon_M |k|)^{-2-\eta}$ ;  $\coth(t\gamma/2) \leq \coth(t_0 m/2)$  and  $\gamma(k) \leq m + |k|$ . Integral comparison of the resulting  $|k|^{-1-\eta}$  tail yields (6.1).

⟨2⟩ 2 The source-phase tail satisfies

$$\|\Phi(J_q) - \Phi(J_q^{(K)})\|_2^2 \leq C_\xi (K/m)^{-\eta}, \quad \|e^{i\Phi(J_q)} - e^{i\Phi(J_q^{(K)})}\|_2 \leq C_\xi (K/m)^{-\eta/2}. \quad (6.2)$$

*Because:* Its variance is the sharp-time thermal covariance paired with the source tail:

$$\sum_{|k| > K} \frac{\coth(t\gamma(k)/2)}{2\gamma(k)} |a_\infty(k)|^2.$$

Use the D4 envelope and integral comparison. The second assertion uses  $|e^{ix} - e^{iy}| \leq |x - y|$ .

⟨2⟩ 3 The shifted action difference obeys

$$\|\mathcal{U}_t^{(K)}(\eta + H_p^{(K)}) - \mathcal{U}_t(\eta + H_p)\|_2 \leq C_\xi [1 + \log(1 + K/m)]^2 (K/m)^{-1/2}. \quad (6.3)$$

⟨3⟩ 1 Expand every shifted Wick power using (3.12) and its quadratic analogue.

⟨3⟩ 2 Terms in which the random Wick chaos is truncated are bounded by (2.17)–(2.19).

⟨3⟩ 3 For deterministic coefficients,

$$\sup_s \frac{1}{2L} \sum_{|k| > K} |\hat{H}_p(s, k)|^2 \leq C_\xi (\varepsilon_M K)^{-1-\eta}, \quad \|H_p^{(K)}\|_\infty, \|H_p\|_\infty \leq C_\xi. \quad (6.4)$$

*Because:* The harmonic time factor has modulus at most one; square the D4 mode envelope and sum its tail. The uniform bound is the absolutely summable  $|k|^{-1-\eta/2}$  envelope.

⟨3⟩ 4 The identity

$$A^{*n} - B^{*n} = \sum_{r=0}^{n-1} A^{*r} * (A - B) * B^{*(n-1-r)} \quad (6.5)$$

and Young/Cauchy–Schwarz bounds turn (6.4) into the required differences of all coefficient power symbols.

*Because:* Only  $n \leq 4$  occurs and the untruncated coefficient sequences have uniform  $\ell^1$  bounds. Each term of (6.5) contains exactly one tail factor, so its  $\ell^2$  norm has the rate in (6.4). Wick

isometry pairs it with the bounded covariance convolution from V2. The slower common rate in (6.3) is therefore valid.

⟨2⟩ 4 For every finite  $p$ ,

$$\sup_K \|e^{-\mathcal{U}_t^{(K)}(\eta+H_p^{(K)})}\|_p < \infty, \quad e^{-\mathcal{U}_t(\eta+H_p)} \in L^p. \quad (6.6)$$

*Because:* Apply (1.4) pointwise with  $u = \eta^{(K)}(s, x) + H_p^{(K)}(s, x)$  and integrate: this gives  $\mathcal{U}_t^{(K)}(\eta + H_p^{(K)}) \geq -Ct\ell_K^2$ , uniformly in the admissible coarse data. Equation (6.3) is the required  $L^2$  increment bound; use  $1 + \log(1 + K/m) \leq C_\varsigma(K/m)^\varsigma$  with  $0 < \varsigma < 1/4$  to retain a positive power. Step ⟨1⟩5 then gives the uniform  $L^p$  bound, and an almost-everywhere convergent subsequence followed by Fatou gives the limiting assertion.

⟨2⟩ 5 This proves ⟨1⟩14. □

⟨1⟩ 15 Prove the continuum periodic partition and marked-Weyl formulas.

⟨2⟩ 1 For  $t \geq t_0$ ,

$$\mathrm{Tr} e^{-tK_L} = Z_0(t) \mathbb{E}_{\mu_{t,L}^{\mathrm{per}}} [e^{-\mathcal{U}_t}] \in (0, \infty). \quad (6.7)$$

*Because:* Equation (3.8) holds at every  $K$ . Its left side converges by (5.15). For the right side, (4.2), (2.16), Hölder, and the uniform exponential bounds give  $e^{-\mathcal{U}_t^{(K)}} \rightarrow e^{-\mathcal{U}_t}$  in  $L^1$ . Finiteness follows. Positivity follows because the exponential is positive a.s.; alternatively Jensen gives  $\mathbb{E}e^{-\mathcal{U}_t} \geq e^{-\mathbb{E}\mathcal{U}_t} > 0$ , and the finite chaos expectation is finite.

⟨2⟩ 2 For the full refined coarse symbol,

$$\frac{\mathrm{Tr}(e^{-tK_L} W_{\mathrm{ct}}(R_M^\infty \xi))}{\mathrm{Tr} e^{-tK_L}} = e^{-S_\infty(H_p)} \frac{\mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta+H_p)}]}{\mathbb{E}[e^{-\mathcal{U}_t}]}. \quad (6.8)$$

⟨3⟩ 1 Apply (3.10) to the symbol and interaction truncated at  $K$ .

⟨3⟩ 2 The Euclidean numerator converges.

*Because:* Split its difference into a phase difference and an exponential difference. Bound the phase part by (6.2) times the uniform  $L^2$  norm in (6.6). Bound the exponential part using (4.2), (6.3), and (6.6). Equations (6.1) and (6.7) handle the prefactor and denominator.

⟨3⟩ 3 The operator numerator converges.

*Because:* Split

$$\mathrm{Tr}[(T_K(t) - T(t))W(\xi_K)] + \mathrm{Tr}[T(t)(W(\xi_K) - W(R_M^\infty \xi))].$$

The first term tends to zero by (5.17) and unitarity. The one-particle symbols  $\xi_K$  converge in norm by the D4 envelope. The Weyl relation and the Fock vacuum overlap

$$\langle W(\alpha)\Omega_0, W(\beta)\Omega_0 \rangle = e^{i\sigma(\alpha,\beta)/2} e^{-\|\alpha-\beta\|^2/4}$$

give strong continuity of  $W(\xi_K)$ . Expanding the trace-class  $T(t)$  in an eigenbasis and using the bound  $2e^{-t\mu_j}$  gives convergence of the second term by dominated convergence. Thus (6.8) follows.

⟨2⟩ 3 This proves ⟨1⟩15. □

## 8 Positivity improvement, simple ground state, and positive gap

⟨1⟩ 16 Prove the continuum ground-state assertions.

⟨2⟩ 1  $T(t)$  is positivity preserving.

*Because:* Each  $T_K(t)$  has a nonnegative Feynman–Kac kernel; strong limits preserve the closed positive cone in  $L^2$ .

⟨2⟩ 2 If  $f, g \geq 0$  are nonzero and bounded, then

$$\langle f, T(t)g \rangle > 0. \quad (7.1)$$

*Because:* Equation (4.5) has a nonnegative integrand. The weight is strictly positive a.s. because  $\mathcal{W}$  is finite a.s. ( $\mathcal{W} \in L^2$ ). If  $A = \{f > 0\}$  and  $B = \{g > 0\}$ , then

$$\mathbb{P}(X_0 \in A, X_t \in B) = \langle \mathbf{1}_A, \Gamma(e^{-t\gamma})\mathbf{1}_B \rangle > 0$$

by (2.3) and Simon’s positivity-improving theorem, since  $\|e^{-t\gamma}\| = e^{-tm} < 1$ . The integrand is strictly positive on an event of positive probability.

⟨2⟩ 3 Equation (7.1) extends to all nonzero  $f, g \geq 0$  in  $L^2$ .

*Because:* Replace each by its bounded truncations  $f \wedge n, g \wedge n$  and use positivity preservation plus  $L^2$  convergence.

⟨2⟩ 4 The largest eigenvalue of the compact positive operator  $T(t)$  is simple and has an a.e. strictly positive eigenvector  $\Omega_L$ .

*Because:* If  $u$  is a top eigenvector, positivity gives  $T(t)|u| \geq |T(t)u| = \|T(t)\||u|$ . The variational characterization forces equality, so  $|u|$  is a nonnegative top eigenvector; (7.1) makes it strictly positive. If a second real top eigenvector  $h$  were orthogonal to it, then both  $h_+$  and  $h_-$  would be nonzero. Positivity improvement gives  $Th_+ > 0$  and  $Th_- > 0$  a.e., hence  $|Th| = |Th_+ - Th_-| < Th_+ + Th_- = T|h|$  a.e. Pairing with  $\Omega_L > 0$  contradicts equality of their top Rayleigh quotients.

⟨2⟩ 5 If  $\mu_0 < \mu_1 \leq \dots$  are the discrete eigenvalues of  $K_L$ ,

$$\gamma_L := \mu_1 - \mu_0 > 0. \quad (7.2)$$

*Because:* Simplicity says the eigenspace at  $\mu_0$  has dimension one; discreteness from Step ⟨1⟩13 prevents another spectral point from accumulating at  $\mu_0$ . Hence the next ordered eigenvalue is strictly larger.

⟨2⟩ 6 The vector functional

$$\omega_{L,\lambda}^{\text{ct}}(A) = \langle \Omega_L, A\Omega_L \rangle \quad (7.3)$$

is a state on the continuum Weyl algebra.

*Because:* Every Weyl operator whose symbol lies in  $H^{-1/2}(\mathbb{T}_L) \oplus iH^{1/2}(\mathbb{T}_L)$  is unitary on the Fock/Gaussian representation. A normalized vector functional is positive and unital; it extends by continuity to the generated  $C^*$ -algebra. The D4 refined coarse symbols lie in this one-particle space by their decay envelope.

⟨2⟩ 7 This proves ⟨1⟩16. □

## 9 V3 conclusion and scope fence

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Steps  $\langle 1 \rangle 9$ – $\langle 1 \rangle 10$  construct the bounded-below self-adjoint  $K_L$  and identify its operator and quadratic form on  $\mathcal{D}$ . Steps  $\langle 1 \rangle 11$ – $\langle 1 \rangle 13$  prove trace class for every positive time, trace-norm cutoff convergence, discrete spectrum, and eigenvalue convergence. Steps  $\langle 1 \rangle 14$ – $\langle 1 \rangle 15$  prove the periodic partition and marked-Weyl formulas. Step  $\langle 1 \rangle 16$  proves a simple strictly positive ground vector and  $\gamma_L > 0$ .

No assertion of essential self-adjointness of  $(H_0 + V_L)|_{\mathcal{D}}$ , and no assertion that  $\mathcal{D}$  is a form core, is made or needed. The source V3’s compressed periodic N2 tail count is repaired explicitly in (2.17)–(2.19).

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